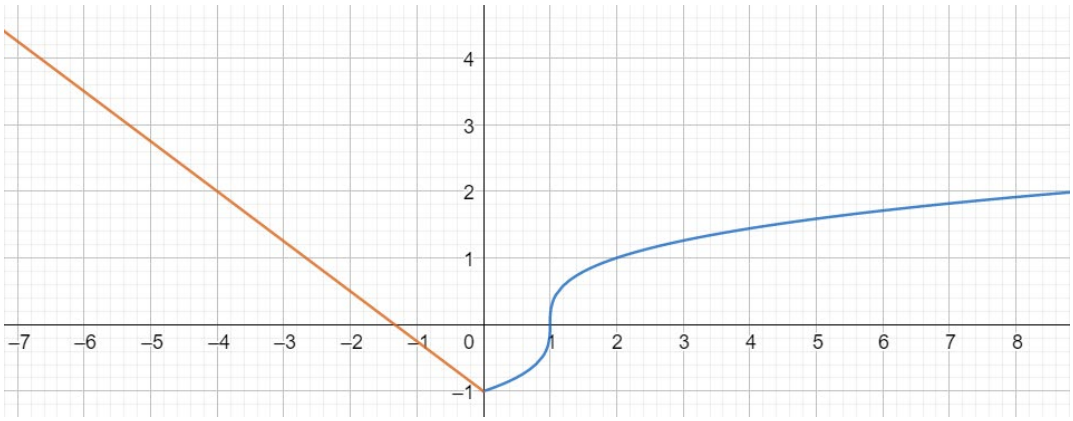
Rajinder Lal
Math 400

Exam 1 Study Guide Solutions

Concept: Find limits graphically.

1. Let $f(x)=\begin{cases}\sqrt[3]{x-1}, \text{ if } x \geq 0 \\ -\frac{3}{4}x-1, \text{ if } x<0\end{cases}$.

(a.) Graph the piecewise function. Find the domain and range of the function.



$Domain(f)=(-\infty,\infty)$ $Range(f)=[-1,\infty)$

(b.) Use the graph of f to find $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$.

$\lim_{x\to 0^-} f(x)=-1$ $\lim_{x\to 0^+} f(x)=-1$

(c.) Find $\lim_{x\to 0} f(x)$, if it exist. If it does not exist, then explain why using appropriate limit notation.

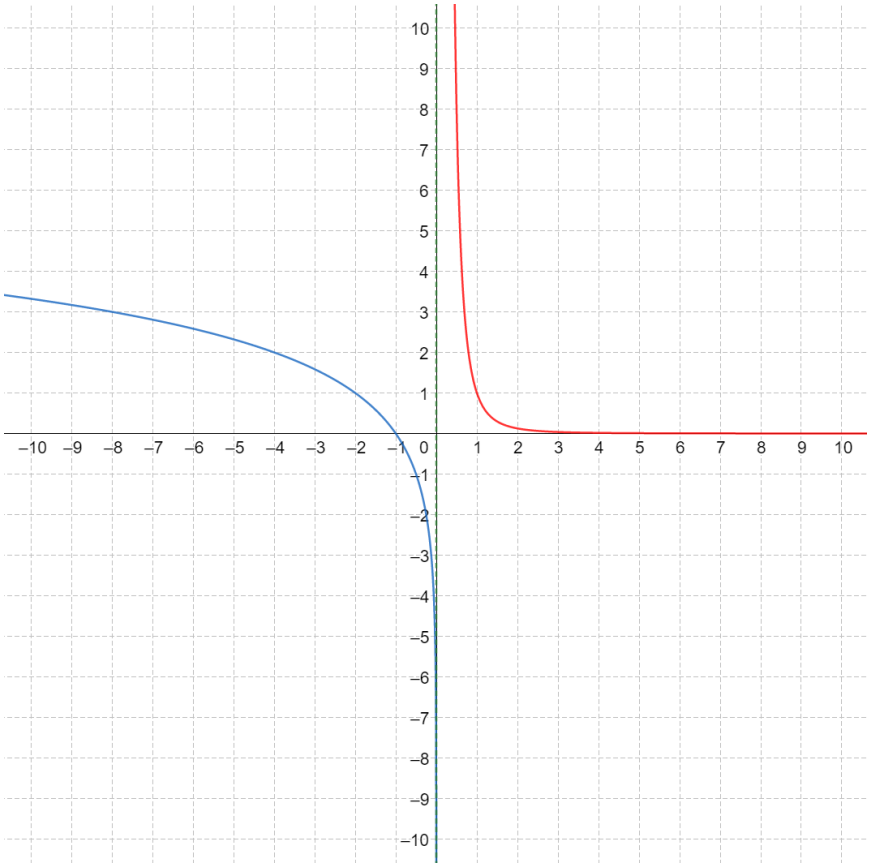
$\lim_{x\to 0} f(x)=-1$

(d.) Find $\lim_{x\to 8} f(x)$, if it exist. If it does not exist, then explain why using appropriate limit notation.

$\lim_{x\to 8} f(x)=\lim_{x\to 8}\sqrt[3]{x-1}=\sqrt[3]{8-1}=\sqrt[3]{7}$

2. Let $f(x)=\begin{cases}\log_2(-x)\ , \text{if } x<0\\ \dfrac{1}{x^3}\ , \text{if } x>0\end{cases}$.

(a.) Graph the piecewise function. Find the domain and range of the function.



Domain$(f)=(-\infty,0)\cup(0,\infty)$ *Range*$(f)=(-\infty,\infty)$

(b.) Find $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$.

$\lim_{x\to 0^-} f(x) = -\infty$ (does not exist)

$\lim_{x\to 0^+} f(x) = \infty$ (does not exist)

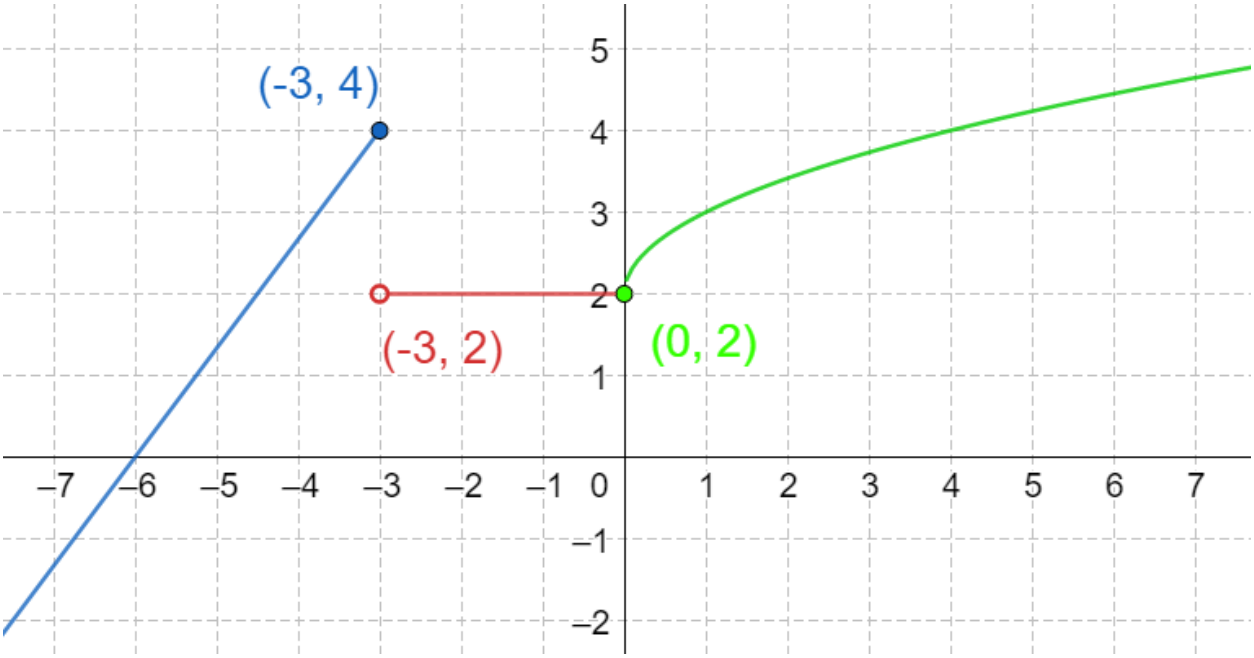
(c.) Find $\lim_{x\to 0} f(x)$, if it exists. If it does not exist, then explain why using appropriate limit notation.

$\lim_{x\to 0} f(x)$ does not exist since

$\lim_{x\to 0^-} f(x) = -\infty$ (does not exist) and $\lim_{x\to 0^+} f(x) = \infty$ (does not exist)

3. Let $f(x)=\begin{cases}\sqrt{x}+2, & \text{if } x\geq 0\\ 2, & \text{if } -3<x<0\\ \frac{4}{3}x+8, & \text{if } x\leq -3\end{cases}$.

(a.) Graph the function f. Find the domain and range of the function.



$Domain(f)=(-\infty,\infty)$ $Range(f)=(-\infty,\infty)$

(b.) Use the graph of f to find $\lim_{x\to 0^-} f(x)$.

$\lim_{x\to 0^-} f(x) = 2$

(c.) Use the graph of f to find $\lim_{x\to 0^+} f(x)$.

$\lim_{x\to 0^+} f(x) = 2$

(d.) Find $\lim_{x\to 0} f(x)$, if it exists. If it does not exist, then explain why using appropriate limit notation.

$\lim_{x\to 0} f(x) = 2$

3. (continued)

(e.) Use the graph of f to find $\lim_{x\to 3^-} f(x)$.

$\lim_{x\to 3^-} f(x) = 4$

(f.) Use the graph of f to find $\lim_{x\to 3^+} f(x)$. (Correction.)

$\lim_{x\to 3^+} f(x) = 2$

(g.) Find $\lim_{x\to 3} f(x)$, if it exists. If it does not exist, then explain why using appropriate limit notation.

$\lim_{x\to 3} f(x)$ does not exist since $\lim_{x\to 3^-} f(x) = 4 \neq 2 = \lim_{x\to 3^+} f(x)$

(h.) Use the graph of f to find $\lim_{x\to 4^-} f(x)$.

$\lim_{x\to 4^-} f(x) = 4$

(i.) Use the graph of f to find $\lim_{x\to 4^+} f(x)$.(Correction.)

$\lim_{x\to 4^+} f(x) = 4$

(j.) Find $\lim_{x\to 4} f(x)$, if it exist. If it does not exist, then explain why using appropriate limit notation.

$\lim_{x\to 4} f(x) = 4$

Concept: Find limits at finite numbers algebraically. These limits have the indeterminate form 0/0.

4. Find the following limits algebraically.

$(a.)\ \lim_{x\to -1}\dfrac{x^4+4x^2-5}{x^3+1}$

$$\lim_{x\to -1}\frac{(x^2+5)(x^2-1)}{(x+1)(x^2-x+1)}=\lim_{x\to -1}\frac{(x^2+5)(x-1)(\boldsymbol{x+1})}{(\boldsymbol{x+1})(x^2-x+1)}=\lim_{x\to -1}\frac{(x^2+5)(x-1)}{x^2-x+1}$$

$$=\frac{((-1)^2+5)((-1)-1)}{(-1)^2-(-1)+1}=\frac{-12}{3}=-4$$

$(b.)\ \lim_{x\to -1}\dfrac{5x^2+3x-14}{15x^2-16x-7}$

$$=\frac{5(-1)^2+3(-1)-14}{15(-1)^2-16(-1)-7}=\frac{-12}{24}=-\frac{1}{2}$$

$$\lim_{x\to \frac{7}{5}}\frac{5x^2+3x-14}{15x^2-16x-7}=\lim_{x\to \frac{7}{5}}\frac{(\boldsymbol{5x-7})(x+2)}{(\boldsymbol{5x-7})(3x+1)}=\lim_{x\to \frac{7}{5}}\frac{x+2}{3x+1}=\frac{\frac{7}{5}+2}{3\left(\frac{7}{5}\right)+1}=\frac{17}{26}$$

$(c.)\ \lim_{h\to 0}\dfrac{\sqrt{12+h}-2\sqrt{3}}{h}$

$$\lim_{h\to 0}\frac{\sqrt{12+h}-2\sqrt{3}}{h}\cdot\frac{\sqrt{12+h}+2\sqrt{3}}{\sqrt{12+h}+2\sqrt{3}}=\lim_{h\to 0}\frac{\left(\sqrt{12+h}\right)^2-\left(2\sqrt{3}\right)^2}{h\left(\sqrt{12+h}+2\sqrt{3}\right)}$$

$$=\lim_{h\to 0}\frac{12+h-12}{h\left(\sqrt{12+h}+2\sqrt{3}\right)}=\lim_{h\to 0}\frac{\boldsymbol{h}}{\boldsymbol{h}\left(\sqrt{12+h}+2\sqrt{3}\right)}=\lim_{h\to 0}\frac{1}{\sqrt{12+h}+2\sqrt{3}}=\frac{1}{\sqrt{12+0}+2\sqrt{3}}$$

$$=\frac{1}{2\sqrt{3}+2\sqrt{3}}=\frac{1}{4\sqrt{3}}$$

4. (continued)

$(d.)\ \lim_{x\to 625}\dfrac{x-625}{25-\sqrt{x}}$

$$\lim_{x\to 625}\frac{x-625}{25-\sqrt{x}}\cdot\frac{25+\sqrt{x}}{25+\sqrt{x}}=\lim_{x\to 625}\frac{(x-625)(25+\sqrt{x})}{625-x}=\lim_{x\to 625}\frac{-(625-x)(25+\sqrt{x})}{625-x}$$

$$=\lim_{x\to 625}-(25+\sqrt{x})=-\left(25+\sqrt{625}\right)=-50$$

$(e.)\ \lim_{h\to 0}\dfrac{\dfrac{1}{(9+h)^2}-\dfrac{1}{9^2}}{h}$

$$\lim_{h\to 0}\frac{\frac{1}{(9+h)^2}-\frac{1}{9^2}}{h}\cdot\frac{9^2(9+h)^2}{9^2(9+h)^2}=\lim_{h\to 0}\frac{9^2-(9+h)^2}{h9^2(9+h)^2}=\lim_{h\to 0}\frac{81-81-18h-h^2}{81h(9+h)^2}$$

$$=\lim_{h\to 0}\frac{-18h-h^2}{81h(9+h)^2}=\lim_{h\to 0}\frac{\boldsymbol{h}(-18-h)}{81\boldsymbol{h}(9+h)^2}=\lim_{h\to 0}\frac{-18-h}{81(9+h)^2}=\frac{-18-0}{81(9+0)^2}=-\frac{18}{81(81)}=-\frac{2}{729}$$

$(f.)\ \lim_{x\to 2}\dfrac{x^4-16}{2-x}$

$$\lim_{x\to 2}\frac{x^4-16}{2-x}=\lim_{x\to 2}\frac{(x^2-4)(x^2+4)}{-(x-2)}=\lim_{x\to 2}\frac{\boldsymbol{(x-2)}(x+2)(x^2+4)}{-\boldsymbol{(x-2)}}$$

$$=\lim_{x\to 2}[-(x+2)(x^2+4)]=-(2+2)(2^2+4)=-32$$

$(g.)\ \lim_{x\to a}\dfrac{x^2-a^2+(x-a)^{100}}{x-a}=\lim_{x\to a}\dfrac{(x-a)(x+a)+(x-a)^{100}}{x-a}$

$$=\lim_{x\to a}\frac{\boldsymbol{(x-a)}[x+a+(x-a)^{99}]}{\boldsymbol{x-a}}=\lim_{x\to a}[x+a+(x-a)^{99}]=a+a+(a-a)^{99}=2a$$

4. (continued)

$(h.)\ \lim_{x\to 1}\left(\frac{x^2}{x^2-5x+4}-\frac{1}{x^2-5x+4}\right)$

$$=\lim_{x\to 1}\frac{x^2-1}{x^2-5x+4}=\lim_{x\to 1}\frac{(x-1)(x+1)}{(x-1)(x-4)}=\lim_{x\to 1}\frac{x+1}{x-4}=\frac{1+1}{1-4}=-\frac{2}{3}$$

5. Find the following limits algebraically.

$(a.)\ \lim_{x\to\pi}\frac{\cos^2 x+3\cos x+2}{\cos x+1}$

$$\lim_{x\to\pi}\frac{\cos^2 x+3\cos x+2}{\cos x+1}=\lim_{x\to\pi}\frac{(\cos x+1)(\cos x+2)}{\cos x+1}=\lim_{x\to\pi}(\cos x+2)=\cos\pi+2=1$$

$(b.)\ \lim_{x\to 0}\frac{\dfrac{\mathbf{1}}{\mathbf{2+\sin x}}-\dfrac{\mathbf{1}}{\mathbf{2}}}{\sin x}$

$$\lim_{x\to 0}\frac{\dfrac{\mathbf{1}}{\mathbf{2+\sin x}}-\dfrac{\mathbf{1}}{\mathbf{2}}}{\sin x}\cdot\frac{2(2+\sin x)}{2(2+\sin x)}=\lim_{x\to 0}\frac{2-(2+\sin x)}{2\sin x\,(2+\sin x)}=\lim_{x\to 0}\frac{-\sin x}{2\sin x\,(2+\sin x)}$$

$$=\lim_{x\to 0}\frac{-1}{2(2+\sin x)}=-\frac{1}{2(2+\sin 0)}=-\frac{1}{4}$$

6. Find the following limit, if it exists. If it does not exist, then explain why using appropriate limit notation.

$$\lim_{x\to 4}\frac{x^2-16}{|x-4|}$$

$$\lim_{x\to 4}\frac{x^2-16}{|x-4|}$$

$$\lim_{x\to 4^+}\frac{x^2-16}{|x-4|}=\lim_{x\to 4^+}\frac{(x-4)(x+4)}{x-4}=\lim_{x\to 4^+}(x+4)=4+4=8$$

$$\lim_{x\to 4^-}\frac{x^2-16}{|x-4|}=\lim_{x\to 4^-}\frac{(x-4)(x+4)}{-(x-4)}=\lim_{x\to 4^-}-(x+4)=-(4+4)=-8$$

$$\lim_{x\to 4^+}\frac{x^2-16}{|x-4|}=8\neq -8=\lim_{x\to 4^-}\frac{x^2-16}{|x-4|}\quad\Rightarrow\quad\lim_{x\to 4}\frac{x^2-16}{|x-4|}\quad does\ not\ exist$$

Concept: Find limits using the limit laws and continuity.

7. Find the following limits.

$$(a.)\lim_{x\to 0}(4x^3-2x^2+10)=4(0)^3-2(0)^2+10=10$$

$$(b.)\lim_{x\to 4}\frac{x-10}{x^3+2x}=\frac{4-10}{4^3+2(4)}=-\frac{6}{72}=-\frac{1}{12}$$

$$(c.)\lim_{x\to 3}\sqrt{\frac{x^2+9x}{x-1}}=\sqrt{\frac{3^2+9(3)}{3-1}}=\sqrt{\frac{36}{2}}=\sqrt{18}=3\sqrt{2}$$

$$(d.)\lim_{x\to \pi}2e^{x/\pi}\cos x=2e^{\pi/\pi}\cos\pi=2e^1(-1)=-2e$$

Concept: Use the Squeeze theorem to find a limit.

8. Use the squeeze theorem to verify that $\lim_{x\to 0} x^4 \sin\left(\frac{1}{x^3}\right) = 0$.

$-1 \le \sin\left(\frac{1}{x^3}\right) \le 1 \quad \Rightarrow \quad -x^4 \le x^4 \sin\left(\frac{1}{x^3}\right) \le x^4$

$\lim_{x\to 0} x^4 = 0 \ \ and \ \ \lim_{x\to 0} -x^4 = 0$

So by the Squeeze Theorem $\lim_{x\to 0} x^4 \sin\left(\frac{1}{x^3}\right) = 0$.

Use the definition of continuity to show that a function is continuous or discontinuous at $x = a$. Find the intervals where a function is continuous.

9. Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), \text{if } x \neq 0 \\ 0, \text{if } x = 0 \end{cases}$. Is f continuous at $x = 0$? Explain.

1.) $f(0) = 0$

2.) We will show that $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$ by using the squeeze theorem.

$-1 \le \sin\left(\frac{1}{x}\right) \le 1$

$x^2(-1) \le x^2 \sin\left(\frac{1}{x}\right) \le x^2(1)$

$-x^2 \le x^2 \sin\left(\frac{1}{x}\right) \le x^2$

$\lim_{x\to 0} x^2 = 0$ and $\lim_{x\to 0} -x^2 = 0$

So by the squeeze theorem $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$

9. (continued)

3.) So $\lim_{x\to 0} f(x) = 0 = f(0)$

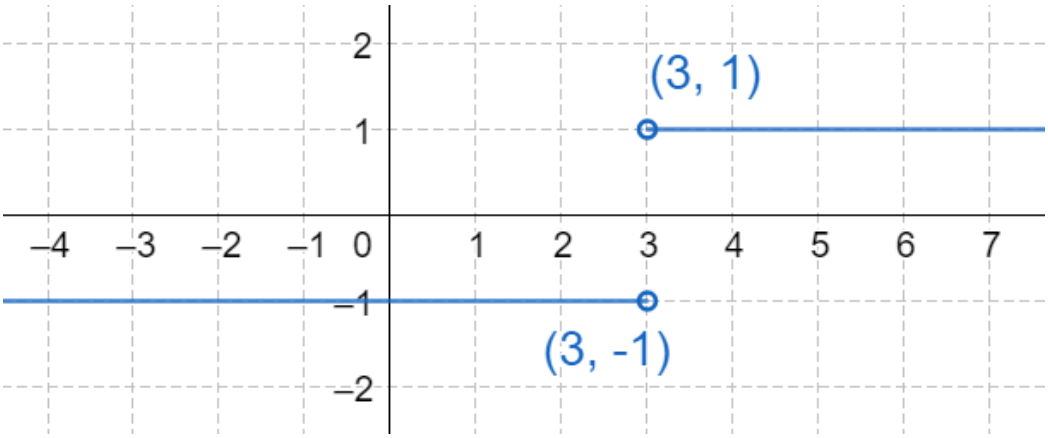
So *f* is continuous at $x = 0$.

10. Let $f(x) = \dfrac{|x-3|}{x-3}$.

(a.) Sketch a graph of *f*.

$$|x-3| = \begin{cases} x-3, \text{ if } x \geq 3 \\ -(x-3), \text{ if } x < 3 \end{cases}$$

$$f(x) = \frac{|x-3|}{x-3} = \begin{cases} \dfrac{x-3}{x-3}, \text{ if } x > 3 \\ \dfrac{-(x-3)}{x-3}, \text{ if } x < 3 \end{cases} = \begin{cases} 1, \text{ if } x > 3 \\ -1, \text{ if } x < 3 \end{cases}$$



(b.) Find all values of x, where *f* is discontinuous. Use the definition of continuity to explain your answer.

f is discontinuous at $x = 3$, since f is undefined at $x = 3$.

Also, $\lim_{x\to 3} f(x)$ does not exist.

$\lim_{x\to 3^-} f(x) = -1 \neq 1 = \lim_{x\to 3^+} f(x) \Rightarrow \lim_{x\to 3} f(x)$ does not exist

11. Find a value of *k* that will make the following function continuous.

$$f(x)=\begin{cases} x^2+3 \text{ , if } x<0 \\ -4x+k, \text{ if } x\geq 0 \end{cases}$$

If $x < 0$, then $f(x) = x^2 + 3$ is continuous on $(-\infty, 0)$.

If $x > 0$, then $f(x) = -4x + k$ is continuous on $(0, \infty)$.

We need to find the value of k that will make f continuous at 0.

$f(0) = -4(0) + k = k$

$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} (x^2 + 3) = 0^2 + 3 = 3$

$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} (-4x + k) = -4(0) + k = k$

In order for f to be continuous at $= 0$, $\lim_{x\to 0^-} f(x) = 3 = k = \lim_{x\to 0^+} f(x)$

So $k = 3$.

If $k = 3$, then f will be continuous on $(-\infty, \infty)$

12. Determine whether the following function is continuous at a. Use the definition of continuity to verify your answer.

$$f(x)=\begin{cases}\dfrac{x^2+x}{x+1}\text{, if } x\neq -1\\ x^2+3x\text{, if } x=-1\end{cases},\quad a=-1$$

1.) $f(-1)=(-1)^2+3(-1)=-2$

2.) $\lim\limits_{x\to -1} f(x)=\lim\limits_{x\to -1}\dfrac{x^2+x}{x+1}=\lim\limits_{x\to -1}\dfrac{x(x+1)}{x+1}=\lim\limits_{x\to -1} x=-1$

3.) $\lim\limits_{x\to -1} f(x)=-1\neq -2=f(-1)$

So f is not continuous at $x=-1$.

13. How would you define $f(2)$ in order to make f continuous at 2? Verify that this satisfies the definition of continuity.

$$f(x)=\frac{x^2-4}{x^2+x-6}$$

n order for f to be continuous at x = 2, we need to have

$\lim\limits_{x\to 2} f(x)=\lim\limits_{x\to 2}\dfrac{x^2-4}{x^2+x-6}=f(2)$.

$$\lim_{x\to 2} f(x)=\lim_{x\to 2}\frac{x^2-4}{x^2+x-6}=\lim_{x\to 2}\frac{(x-2)(x+2)}{(x-2)(x+3)}=\lim_{x\to 2}\frac{x+2}{x+3}=\frac{2+2}{2+3}=\frac{4}{5}$$

So I would define $f(2)=\dfrac{4}{5}$.

14. Determine the values of x at which the following function f has discontinuities.

$$f(x)=\frac{1+\sin x}{\cos x}$$

The functions $y = 1 + \sin x$ and $y = \cos x$ are continuous on their domain, $(-\infty, \infty)$.

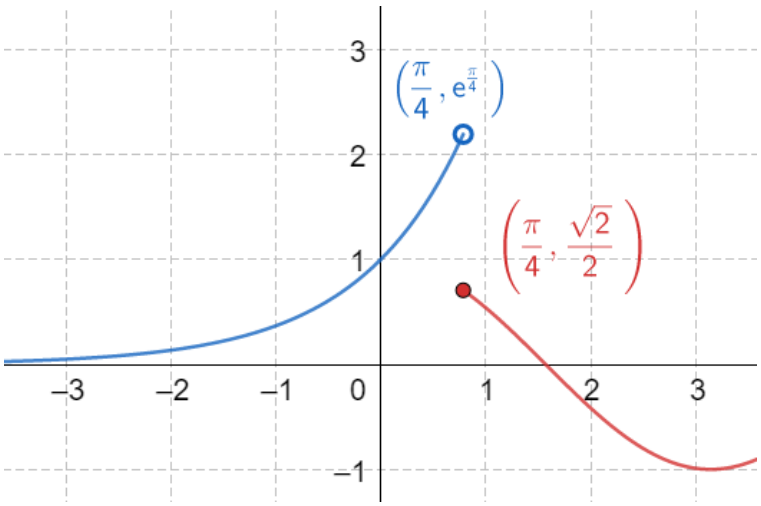
$f(x)=\frac{1+\sin x}{\cos x}$ will have discontinuities when $\cos x = 0$.

$\cos x = 0$ when $x=\frac{\pi}{2}+\pi k$, where k is an integer.

So $f(x)=\frac{1+\sin x}{\cos x}$ has discontinuities when $x=\frac{\pi}{2}+\pi k$, where k is an integer.

15. Sketch the graph of the function. Use the definition of continuity to explain why the function is discontinuous at $x=\frac{\pi}{4}$.

$$f(x)=\begin{cases}\cos x, & \text{if } x\geq\frac{\pi}{4}\\ e^x, & \text{if } x<\frac{\pi}{4}\end{cases}$$



1.) $f\left(\frac{\pi}{4}\right)=\cos\left(\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}$

2.) $\lim\limits_{x\to\frac{\pi}{4}^+} f(x)=\lim\limits_{x\to\frac{\pi}{4}^+}\cos x=\cos\left(\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}$

$\lim\limits_{x\to\frac{\pi}{4}^-} f(x)=\lim\limits_{x\to\frac{\pi}{4}^-} e^x=e^{\frac{\pi}{4}}$

$\lim\limits_{x\to\frac{\pi}{4}^+} f(x)\neq\lim\limits_{x\to\frac{\pi}{4}^-} f(x)\Rightarrow\lim\limits_{x\to\frac{\pi}{4}} f(x)$ does not exist. $\Rightarrow f$ is discontinuous at $x=\frac{\pi}{4}$.

Concept: Find limits at infinity algebraically. These limits have the indeterminate form $\frac{\infty}{\infty}$ or $\infty - \infty$.

16. Find the following limits algebraically.

(a.) $\lim\limits_{x\to\infty} \dfrac{17x^4 - 9x + 1}{5x^4 + 9x^2 + 10x}$

$$\lim_{x\to\infty} \frac{17x^4 - 9x + 1}{5x^4 + 9x^2 + 10x} \cdot \frac{\frac{1}{x^4}}{\frac{1}{x^4}} = \lim_{x\to\infty} \frac{\frac{17x^4}{x^4} - \frac{9x}{x^4} + \frac{1}{x^4}}{\frac{5x^4}{x^4} + \frac{9x^2}{x^4} + \frac{10x}{x^4}} = \lim_{x\to\infty} \frac{17 - \frac{9}{x^3} + \frac{1}{x^4}}{5 + \frac{9}{x^2} + \frac{10}{x^3}} = \frac{17 - 0 + 0}{5 + 0 + 0} = \frac{17}{5}$$

(b.) $\lim\limits_{x\to-\infty} \dfrac{-4x^5 + 3x^2 - 8x + 1}{1 - 16x^2}$

$$\lim_{x\to-\infty} \frac{-4x^5 + 3x^2 - 8x + 1}{1 - 16x^2} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} = \lim_{x\to-\infty} \frac{\frac{-4x^5}{x^2} + \frac{3x^2}{x^2} - \frac{8x}{x^2} + \frac{1}{x^2}}{\frac{1}{x^2} - \frac{16x^2}{x^2}} = \lim_{x\to-\infty} \frac{-4x^3 + 3 - \frac{8}{x} + \frac{1}{x^2}}{\frac{1}{x^2} - 16} = -\infty$$

17. Find the limit algebraically.

(a.) $\lim\limits_{x\to\infty} \dfrac{3\cdot 2^{3x} + 2^x}{2^{7x} - 11\cdot 2^x}$

$$\lim_{x\to\infty} \frac{3\cdot 2^{3x} + 2^x}{2^{7x} - 11\cdot 2^x} \cdot \frac{\frac{1}{2^{7x}}}{\frac{1}{2^{7x}}} = \lim_{x\to\infty} \frac{\frac{3\cdot 2^{3x}}{2^{7x}} + \frac{2^x}{2^{7x}}}{\frac{2^{7x}}{2^{7x}} - \frac{11\cdot 2^x}{2^{7x}}} = \lim_{x\to\infty} \frac{\frac{3}{2^{4x}} + \frac{2}{2^{6x}}}{1 - \frac{11}{2^{6x}}} = \frac{0 + 0}{1 - 0} = 0$$

(b.) $\lim\limits_{x\to\infty} \dfrac{e^{4x} + e^{-4x}}{e^{4x} - e^{-4x}}$

$$\lim_{x\to\infty} \frac{e^{4x} + e^{-4x}}{e^{4x} - e^{-4x}} = \lim_{x\to\infty} \frac{e^{4x} + \frac{1}{e^{4x}}}{e^{4x} - \frac{1}{e^{4x}}} \cdot \frac{\frac{1}{e^{4x}}}{\frac{1}{e^{4x}}} = \lim_{x\to\infty} \frac{\frac{e^{4x}}{e^{4x}} + \frac{1}{e^{4x}}\frac{1}{e^{4x}}}{\frac{e^{4x}}{e^{4x}} - \frac{1}{e^{4x}}\frac{1}{e^{4x}}} = \lim_{x\to\infty} \frac{1 + \frac{1}{e^{8x}}}{1 - \frac{1}{e^{8x}}} = \frac{1 + 0}{1 - 0} = 1$$

18. Determine the end behavior of the following function. You will need to find $\lim_{x\to\infty} h(x)$ and $\lim_{x\to-\infty} h(x)$ algebraically. Indicate the equation(s) of any horizontal asymptote(s).

$$h(x)=\frac{\sqrt{3x^{18}+x^{10}+4}}{-7x^{9}+4x^{2}-9x}$$

$$\lim_{x\to\infty}\frac{\sqrt{3x^{18}+x^{10}+4}}{-7x^{9}+4x^{2}-9x}\cdot\frac{\frac{1}{\sqrt{x^{18}}}}{\frac{1}{x^{9}}}=\lim_{x\to\infty}\frac{\sqrt{3+\frac{1}{x^{8}}+\frac{4}{x^{18}}}}{-7+\frac{4}{x^{7}}-\frac{9}{x^{8}}}$$

$\sqrt{3+\frac{1}{x^{8}}+\frac{4}{x^{18}}}\to\sqrt{3+0+0}=\sqrt{3}$ (inside $\to 0$, $\to 0$); $-7+\frac{4}{x^{7}}-\frac{9}{x^{8}}\to -7+0-0=-7$ ($\frac{4}{x^{7}}\to 0$, $\frac{9}{x^{8}}\to 0$)

$$=\frac{\sqrt{3}}{-7}\Rightarrow y=\frac{\sqrt{3}}{-7}\text{ is a H.A.}$$

$$\lim_{x\to-\infty}\frac{\sqrt{3x^{18}+x^{10}+4}}{-7x^{9}+4x^{2}-9x}\cdot\frac{\frac{1}{\sqrt{x^{18}}}}{\frac{1}{-x^{9}}}=\lim_{x\to-\infty}\frac{\sqrt{3+\frac{1}{x^{8}}+\frac{4}{x^{18}}}}{7-\frac{4}{x^{7}}+\frac{9}{x^{8}}}$$

$\sqrt{3+\frac{1}{x^{8}}+\frac{4}{x^{18}}}\to\sqrt{3+0+0}=\sqrt{3}$ (inside $\to 0$, $\to 0$); $7-\frac{4}{x^{7}}+\frac{9}{x^{8}}\to 7-0+0=7$ ($\frac{4}{x^{7}}\to 0$, $\frac{9}{x^{8}}\to 0$)

$$=\frac{\sqrt{3}}{7}\Rightarrow y=\frac{\sqrt{3}}{7}\text{ is a H.A.}$$

19. Find the limit algebraically.

$$\lim_{x\to\infty}\left(\sqrt{3x^2+8x+6}-\sqrt{3x^2+3x+1}\right)\cdot\frac{\sqrt{3x^2+8x+6}+\sqrt{3x^2+3x+1}}{\sqrt{3x^2+8x+6}+\sqrt{3x^2+3x+1}}$$

$$=\lim_{x\to\infty}\frac{(3x^2+8x+6)-(3x^2+3x+1)}{\sqrt{3x^2+8x+6}+\sqrt{3x^2+3x+1}}$$

$$=\lim_{x\to\infty}\frac{5x+5}{\sqrt{3x^2+8x+6}+\sqrt{3x^2+3x+1}}\cdot\frac{\frac{1}{x}}{\frac{1}{\sqrt{x^2}}}$$

$$=\lim_{x\to\infty}\frac{5+\frac{5}{x}\to 0}{\sqrt{3+\frac{8}{x}+\frac{6}{x^2}}+\sqrt{3+\frac{3}{x}+\frac{1}{x^2}}}$$

(each $\frac{8}{x}$, $\frac{6}{x^2}$, $\frac{3}{x}$, $\frac{1}{x^2}$ $\to 0$)

$$=\frac{5}{\sqrt{3}+\sqrt{3}}$$

$$=\boxed{\frac{5}{2\sqrt{3}}}$$

Concept: Find limits at infinity analytically.

20. Find the following limits analytically.

(a.) $\lim_{x\to-\infty} \arctan\left(e^{|2x-1|}\right)$

$\lim_{x\to-\infty} |2x-1| = \infty$

$\Rightarrow \lim_{x\to-\infty} e^{|2x-1|} = \infty$

$\Rightarrow \lim_{x\to-\infty} \arctan\left(e^{|2x-1|}\right) = \boxed{\frac{\pi}{2}}$

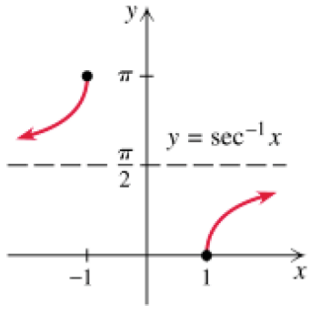
(b.) $\lim_{x\to\infty} \sin\left[\operatorname{arcsec}\left(\frac{x-4}{\sqrt{x}-2}\right)\right]$

(b.) $\lim_{x\to\infty} \sin\left[\operatorname{arcsec}\left(\frac{x-4}{\sqrt{x}-2}\right)\right]$

$\lim_{x\to\infty} \frac{x-4}{\sqrt{x}-2} \cdot \frac{\frac{1}{\sqrt{x}}}{\frac{1}{\sqrt{x}}} = \lim_{x\to\infty} \frac{\sqrt{x} - \frac{4}{\sqrt{x}}}{1 - \frac{2}{\sqrt{x}}} = \infty$ ($\sqrt{x} \to \infty$, $\frac{4}{\sqrt{x}} \to 0$, $1 \to 1$, $\frac{2}{\sqrt{x}} \to 0$)

$\Rightarrow \lim_{x\to\infty} \operatorname{arcsec}\left(\frac{x-4}{\sqrt{x}-2}\right) = \frac{\pi}{2}$

$\Rightarrow \lim_{x\to\infty} \sin\left[\operatorname{arcsec}\left(\frac{x-4}{\sqrt{x}-2}\right)\right] = \boxed{1}$ ($\operatorname{arcsec}\left(\frac{x-4}{\sqrt{x}-2}\right) \to \frac{\pi}{2}$)



Domain $(-\infty, -1] \cup [1, \infty)$

Range $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

(c.) $\lim_{x\to\infty} \ln(\ln(\ln x))$

$x \to \infty \Rightarrow \ln(x) \to \infty \Rightarrow \ln(\ln(x)) \to \infty \Rightarrow \ln(\ln(\ln(x))) \to \infty$

Concept: Find limits at a finite number analytically.

21. Find the following limits analytically. If the limit is ∞ or $-\infty$, then state this.

$(a.)\ \lim_{x\to 1^-} \ln(\cos^{-1} x) = -\infty$

$x \to 1^- \Rightarrow \cos^{-1} x \to 0^+ \Rightarrow \ln(\cos^{-1} x) \to -\infty$

$(b.)\ \lim_{x\to 8^-} \dfrac{x+8}{(x-8)^2(2x+3)}$

$x \to 8^- \Rightarrow x + 8 \to 16$

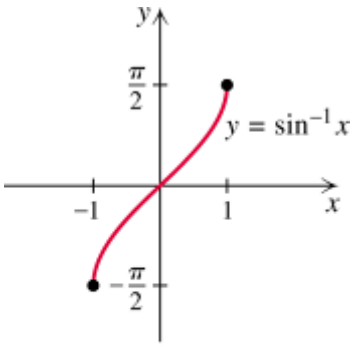
$x \to 8^- \Rightarrow x - 8 \to 0^- \Rightarrow (x-8)^2 \to 0^+$

$x \to 8^- \Rightarrow 2x + 3 \to 19$

We put this all together and get $\lim_{x\to 8^-} \frac{x+8}{(x-8)^2(2x+3)} = \infty$

21. (continued)

$(c.)\ \lim_{x\to 2} \arcsin\left(\dfrac{2-x}{4\left(\sqrt{2}-\sqrt{x}\right)}\right) = \dfrac{\pi}{4}$



Domain $[-1, 1]$
Range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$\lim_{x\to 2} \frac{2-x}{4\left(\sqrt{2}-\sqrt{x}\right)} = \lim_{x\to 2} \frac{2-x}{4\left(\sqrt{2}-\sqrt{x}\right)} \cdot \frac{\sqrt{2}+\sqrt{x}}{\sqrt{2}+\sqrt{x}} = \lim_{x\to 2} \frac{(2-x)(\sqrt{2}+\sqrt{x})}{4\left(\left(\sqrt{2}\right)^2-\left(\sqrt{x}\right)^2\right)}$$

$$= \lim_{x\to 2} \frac{\mathbf{(2-x)}(\sqrt{2}+\sqrt{x})}{4\mathbf{(2-x)}} = \lim_{x\to 2} \frac{\sqrt{2}+\sqrt{x}}{4} = \frac{\sqrt{2}+\sqrt{2}}{4} = \frac{2\sqrt{2}}{4} = \frac{\sqrt{2}}{2}$$

$$x \to 2 \Rightarrow \frac{2-x}{4\left(\sqrt{2}-\sqrt{x}\right)} \to \frac{\sqrt{2}}{2} \Rightarrow \arcsin\left(\frac{2-x}{4\left(\sqrt{2}-\sqrt{x}\right)}\right) \to \arcsin\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{4}$$

$(d.)\ \lim_{x\to\pi^+} \dfrac{1}{x^2}\csc(x-\pi)$

$$x \to \pi^+ \Rightarrow x-\pi \to 0^+ \Rightarrow \csc(x-\pi) \to \infty$$

$$x \to \pi^+ \Rightarrow \frac{1}{x^2} \to \frac{1}{\pi^2}$$

We put this together and get $\lim_{x\to\pi^+} \frac{1}{x^2}\csc(x-\pi) = \infty$.

Concept: Use the Precise Definition of a limit to prove a limit.

22. Prove the statement by using the ε, δ definition of a limit.

$\lim_{x\to 2}(-14x+10) = -18$

Proof: Let $f(x) = -14x + 10$

Let $\varepsilon > 0$.

Choose $\delta = \frac{\varepsilon}{14}$.

Suppose $0 < |x-2| < \delta = \frac{\varepsilon}{14}$.

Then $|-14x+10-(-18)| = |-14x+28| = |-14(x-2)| = |-14||x-2| = 14|x-2| < 14 \cdot \frac{\varepsilon}{14} = \varepsilon$

Therefore, $\lim_{x\to 2}(-14x+10) = -18$.

23. Prove the statement by using the ε, δ definition of a limit.

$$\lim_{x\to -\frac{1}{3}} \frac{9x^2-1}{3x+1} = -2$$

$$Let\ f(x) = \frac{9x^2-1}{3x+1}$$

$$f(x) = \frac{9x^2-1}{3x+1} = \frac{(3x-1)(3x+1)}{3x+1} = 3x-1 \quad if \quad x \neq -\frac{1}{3}$$

Proof: Let $\varepsilon > 0$.

Choose $\delta = \frac{\varepsilon}{3}$.

Suppose $0 < \left|x+\frac{1}{3}\right| < \delta = \frac{\varepsilon}{3}$.

Then $|(3x-1)-(-2)| = |3x+1| = \left|3\left(x+\frac{1}{3}\right)\right| = |3|\left|x+\frac{1}{3}\right| = 3\left|x+\frac{1}{3}\right| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon$

Therefore $\lim_{x\to -\frac{1}{3}} \frac{9x^2-1}{3x+1} = -2$.

24. Use the precise definition of an infinite limit to prove the following.

$$\lim_{x\to 9} \frac{1}{(x-9)^2} = \infty$$

Proof: Let $f(x) = \frac{1}{(x-9)^2}$

Let $M > 0$.

Choose $\delta = \frac{1}{\sqrt{M}}$.

Suppose $0 < |x-1| < \delta = \frac{1}{\sqrt{M}}$.

$$f(x) = \frac{1}{(x-9)^2} = \frac{1}{|x-1|^2} > \frac{1}{\left(\frac{1}{\sqrt{M}}\right)^2} = \frac{1}{\frac{1}{M}} = M$$

Therefore $\lim_{x\to 9} \frac{1}{(x-9)^2} = \infty$.